INVESTIGATION OF A RANSOMWARE ATTACK CAUSED BY PHISHING ON A UNIVERSITY LEARNING MANAGEMENT SYSTEM

REXMI STATE UNIVERSITY (RSU)

Track: Cybersecurity Incident Investigation

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STUDENT ID: WTS/2026/169


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CodeHunters has breached LMS.
A WARNING
If the school IS interested in preventing the release of their data, please contact us privately at TOX to negotiate a settlement.
You have till the end of the day by 12 September 2026 before everything is leaked.
LMS still has until EOD 12 September 2026 to contact us.
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BACKGROUND AND CONTEXT

- Higher education institutions and universities increasingly rely on Learning Management Systems (LMS) in the delivery of coursework, assignment, students grading, and communication.

- Combination of many users within the institution and valuable data stored in their systems makes them liable to attacks such as ransomware attack.

- Phishing attack is still the most common point of entry for this type of cyber-attack since it utilizes human weakness in decision making rather than any system vulnerability making even well-protected networks exposed.



THE PROBLEM

The LMS of Rexmi State University (RSU) was exposed to an attack chain from phishing to ransomware that caused the total loss of access to learning materials, submission of assignments, grades, and course communication.

- Affected the entire RSU student and staff population
- Occurred near the end of the semester, when coursework and grading are most critical
- Network/Online backups were also destroyed

This happened after a junior member of the staff opened a link in an email claiming to be an important IT notice, granting the attacker access to the university's network where encryption of the LMS was done overnight.
This incident did not occur because of any single mistake but because of controls that were lacking when they were each required at their exact moment

```
20:45:02 Email: Message delivered to amonu@rsu.edu, subject "URGENT: IT Account Verification Required"
20:45:47 Webserver: GET /verify-account.html from 41.203.18.22, response 200
20:45:52 Auth: Successful login for user amonu from 192.168.14.30
20:47:10 System: New scheduled task created on WKSTN-114 by user amonu
20:47:15 System: Registry Run key modified on WKSTN-114
20:47:22 Firewall: ALLOWED outbound connection to 185.9.2.1, port 4444
20:48:03 DNS: query for a7fk29dksla02mksd.attacker-cdn.net
20:52:00 System: Windows Event Log service stopped on WKSTN-114
21:30:11 Auth: Successful login for user amonu to DC-01 (Domain Controller)
21:30:18 System: New account added to Domain Admins group on DC-01
21:45:02 Auth: Successful login for user amonu to LMS-FileServer-1
21:45:40 Auth: Successful login for user amonu to LMS-FileServer-2
21:46:05 Auth: Successful login for user boateng to Backup-Server-1
21:46:30 System: Remote service PSEXESVC installed on LMS-FileServer-1
22:15:00 SIEM: Alert - Anomalous account activity for user amonu (multiple hosts, 45 min window)
23:00:04 System: Shadow copies deleted on LMS-FileServer-1
23:00:11 System: Shadow copies deleted on LMS-FileServer-2
23:00:47 Backup: Backup catalog purged on Backup-Server-1
23:02:15 System: Windows Server Backup service stopped on Backup-Server-1
23:31:00 System: Mass file rename operation detected on LMS-FileServer-1 (.locked extension)
00:15:00 System: Mass file rename operation detected on LMS-FileServer-2 (.locked extension)
05:00:00 System: Encryption process completed on LMS-FileServer-1 and LMS-FileServer-2
06:12:00 Webserver: GET /lms/dashboard from 192.168.22.9, response 500 (file not found)
06:25:00 Webserver: GET /lms/coursework from 192.168.22.14, response 500 (file not found)
08:00:00 System: Ransom note file detected: README_DECRYPT.txt on LMS-FileServer-1, LMS-FileServer-2, Backup-Server-1
```

Urgent Account Verification Required

Summarize

Reply | Reply All | Forward

N

Rexmi State University <noreply@notify.rssu.edu>

To Sarah Amonu <amonu@rsu.edu>

Sun 06/09/2026 4:20 pm

Hi Sarah Amonu,

We have detected suspicious activity on your account and require your immediate attention. In order to maintain the security of your account, we kindly request you to verify your account details by clicking on the link below:

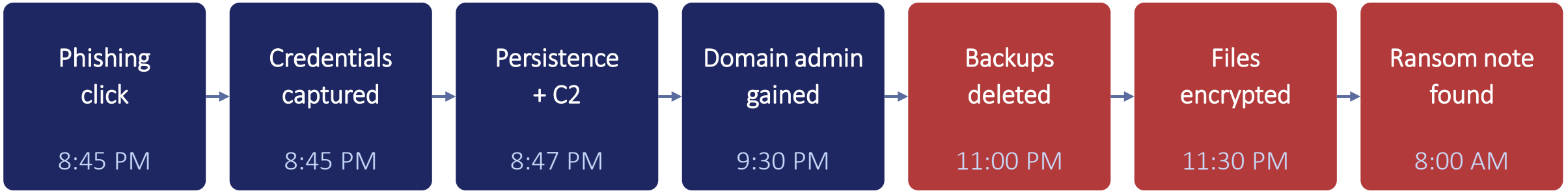
Verify now!

Failure to complete the verification process within 24 hours will result in temporary account suspension. We apologize for the inconvenience and appreciate your prompt response.

Best regards,
IT Team
Rexmi State University

HOW IT WORKED

The attack chain, reconstructed from log evidence and mapped to MITRE ATT&CK:



Each stage succeeded because one specific control was missing: email filtering, SSL inspection, patching, detection tuning, or backup isolation.

ATT&CK coverage:

Initial Access (**Phishing -T1566**) → Credential Access (**Steal Web Session Cookie -T1539**) → Persistence (**Boot AutoStart Execution -T1547**) → C2 (**Application Layer Protocol -T1071)** → Privilege Escalation (**Exploitation for Privilege Escalation -T1068**) → Lateral Movement (**Remote Services -T1021**) → Impact (**Inhibit System Recovery - T1490**, **Data Encrypted for Impact -T1486**)

Common Vulnerability Scoring System Version 4.0 Calculator

CVSS:4.0/AV:N/AC:L/AT:N/PR:L/UI:N/VC:H/VI:H/VA:H/SC:N/SI:N/SA:N Reset

CVSS v4.0 Score: 8.7 / High

Base Metrics

Exploitability Metrics

Attack Vector (AV):	**Network (N)**	Adjacent (A)	Local (L)	Physical (P)
Attack Complexity (AC):	**Low (L)**	High (H)		
Attack Requirements (AT):	**None (N)**	Present (P)		
Privileges Required (PR):	None (N)	**Low (L)**	High (H)	
User Interaction (UI):	**None (N)**	Passive (P)	Active (A)	

Vulnerable System Impact Metrics

Confidentiality (VC):	**High (H)**	Low (L)	None (N)
Integrity (VI):	**High (H)**	Low (L)	None (N)
Availability (VA):	**High (H)**	Low (L)	None (N)

RESEARCH & INSIGHTS

Investigating the incident's logs and timeline surfaced five specific gaps and not a single point of failure, but an unbroken chain of missing controls.

No email filtering

The phishing message and credential-harvesting link reached the inbox unblocked.

No SSL/TLS inspection

Encrypted command-and-control traffic passed through as unreadable, unflagged data.

Unpatched domain controller

CVSS 8.7 (Critical) — a single flaw handed the attacker full domain admin rights.

Detection tuned too late

The SIEM alert fired only after admin access was already secured.

Backups not isolated

Shadow copies and backup catalogs were reachable and destroyed from the same network.

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IMPACT

- The attack affected availability of the LMS. Between 11:30 PM and the discovery of the ransom note at 8:00 AM, there were absolutely no opportunities for any student and lecturer to access the LMS.

- The Integrity triage was affected since files were not only deleted or hidden, but they were also modified (encrypted) as well as the user interface showing the ransom note.

- The attack also affected the Confidentiality of the LMS because the attackers should not have been able to access the LMS data but they got access to the data and encrypted them.



THE SOLUTION

With the response action to the incident, the NIST Incident Response Framework (IRF) was used by RSU.

- **Preparation**

The preparation stage is where the university was supposed to make sufficient preparation to await any future incident that may come but RSU failed to make appropriate preparation.

- **Detection and Analysis**

It took some time for the SIEM system to produce an alert around 10:15 PM, but even after that, the attacker had already acquired access to domain admin privileges.

Four phases

1. Preparation
2. Detection & Analysis
3. Containment, Eradication and Recovery
4. Post-incident Activity

- **Containment, Eradication and Recovery**

After the investigation, the first step the IT team took was the containment and format of the infected systems, a full reset of all the accounts credentials and modification of the firewall rules, IDS & IPS.

The recovery involved the use of the offline backup that had been maintained by RSU as a separate entity and hence was not affected by the attack. However, since RSU offline backups were not daily but rather monthly, the offline backup used for recovery.

- **Post-incident Activity**

Post-incident analysis was performed as the LMS was brought back online, and all the five gaps that contributed to the root cause identified and resolved.

Five fixes

1. Email filtering + request verification
2. SSL/TLS traffic inspection
3. Domain controller patch management
4. Earlier-tuned SIEM detection
5. Backup segmentation

AI USAGE REPORT

Claude AI supported with analysis and drafting of the project which was reviewed and changed to my preference. Google Gemini was also used to generate images for this presentation

Log dataset generation

Claude AI helped generate a realistic Auth / Firewall / System / Webserver log set from the established timeline. I reviewed it and I personally made changes to it to fit my preference

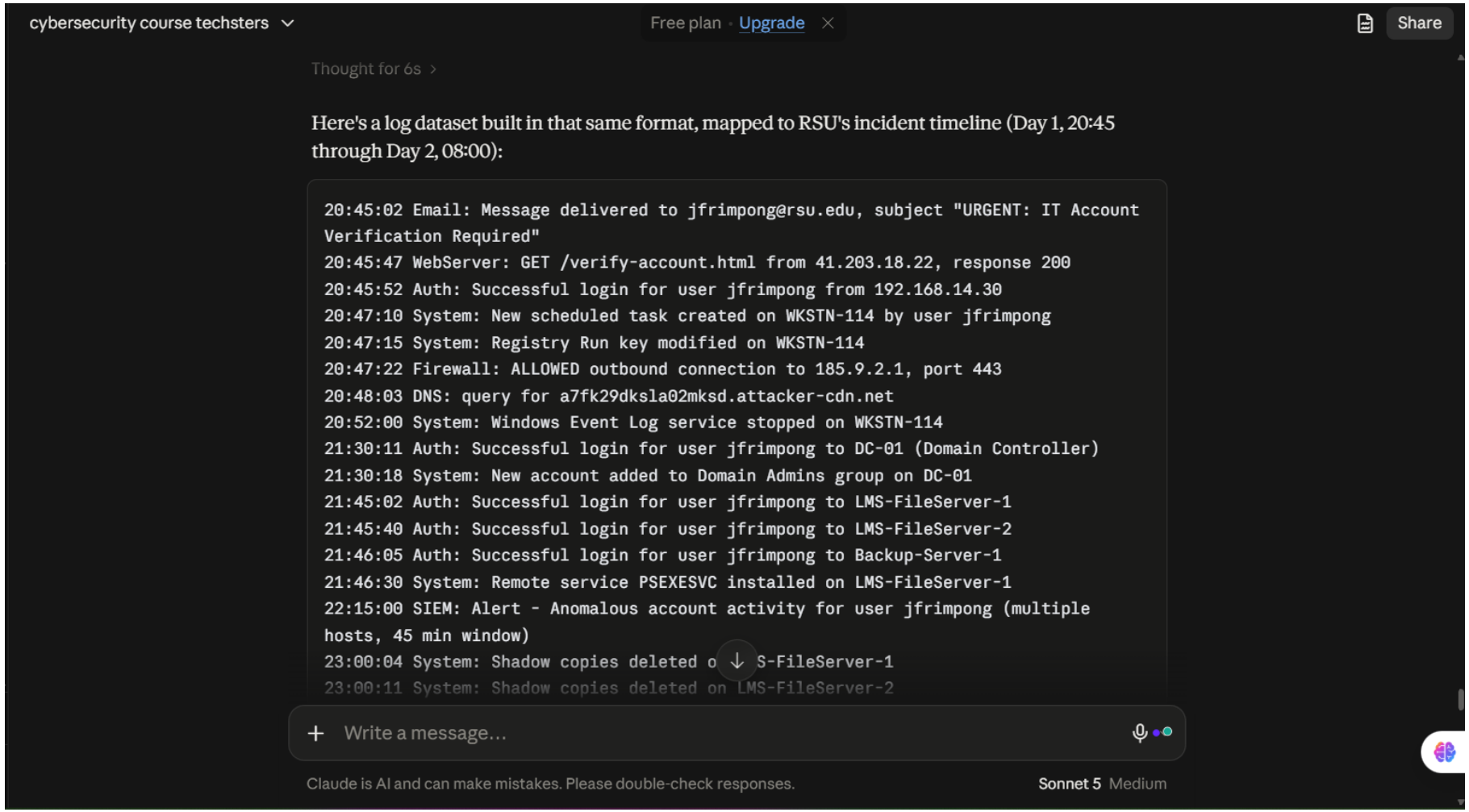
MITRE ATT&CK mapping

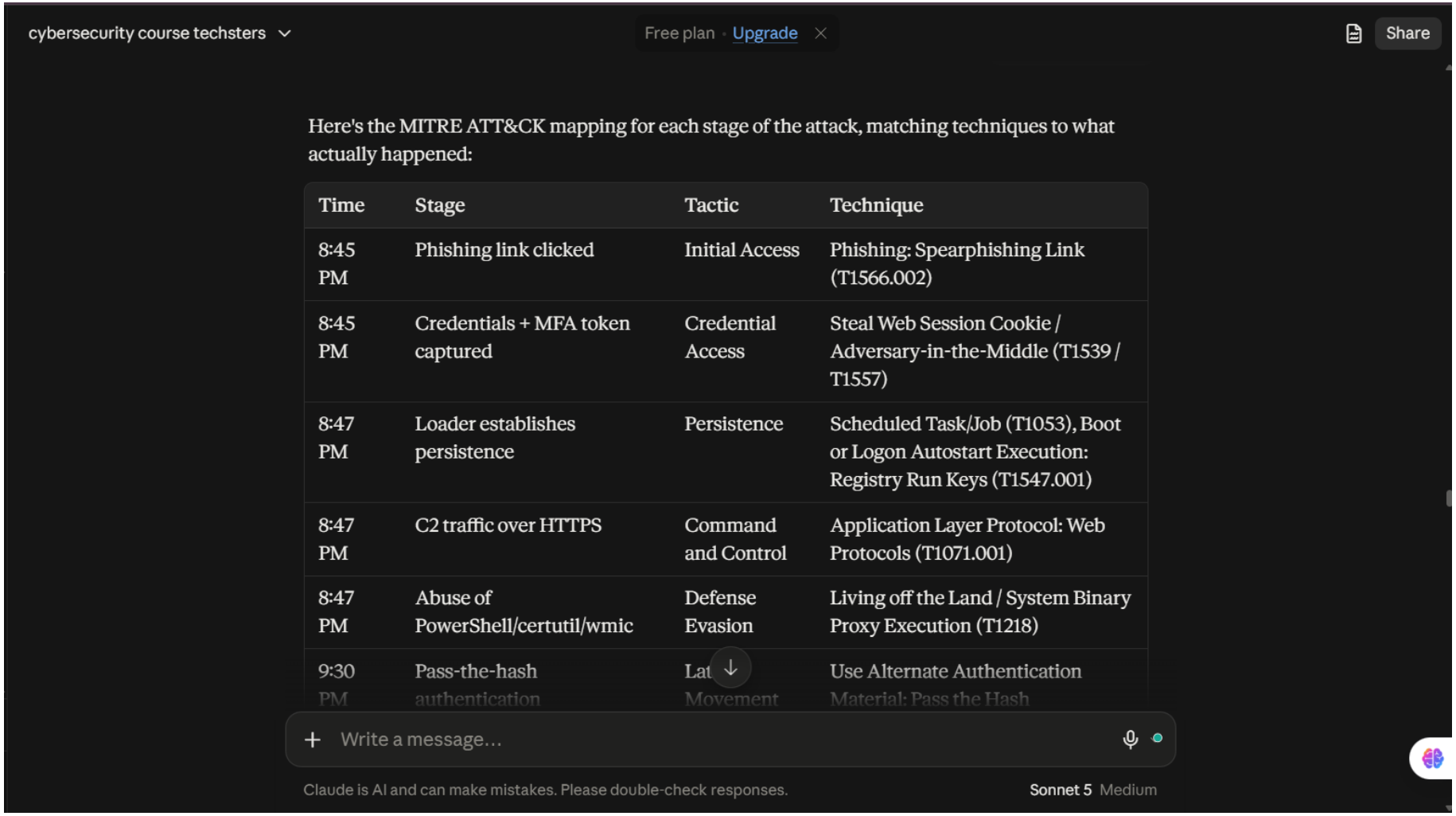
Given the incident timeline, Claude suggested matching techniques. Each suggestion was checked and corrected against MITRE's official descriptions before being used.

Drafting And Summarization

Claude AI was used to draft and summarize the problem statement, background & context and the conclusion of the problem statement report

Gemini AI was also used to generate the image used in the preparation of this PowerPoint slides.


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Thought for 6s
Here's a log dataset built in that same format, mapped to RSU's incident timeline (Day 1, 20:45 through Day 2, 08:00):
20:45:02 Email: Message delivered to jfrimpong@rsu.edu, subject "URGENT: IT Account Verification Required"
20:45:47 WebServer: GET /verify-account.html from 41.203.18.22, response 200
20:45:52 Auth: Successful login for user jfrimpong from 192.168.14.30
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Write a message...
Claude is AI and can make mistakes. Please double-check responses.
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Share
Here's the MITRE ATT&CK mapping for each stage of the attack, matching techniques to what actually happened:
Time
Stage
Tactic
Technique
8:45 PM
Phishing link clicked
Initial Access
Phishing: Spearphishing Link (T1566.002)
8:45 PM
Credentials + MFA token captured
Credential Access
Steal Web Session Cookie / Adversary-in-the-Middle (T1539 / T1557)
8:47 PM
Loader establishes persistence
Persistence
Scheduled Task/Job (T1053), Boot or Logon Autostart Execution: Registry Run Keys (T1547.001)
8:47 PM
C2 traffic over HTTPS
Command and Control
Application Layer Protocol: Web Protocols (T1071.001)
8:47 PM
Abuse of PowerShell/certutil/wmic
Defense Evasion
Living off the Land / System Binary Proxy Execution (T1218)
9:30 PM
Pass-the-hash authentication
Lat Movement
Use Alternate Authentication Material: Pass the Hash
Write a message...
Claude is AI and can make mistakes. Please double-check responses.
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State University (RSU) and remove the board from the back and the innovate & collaborate Univ of Kumasi

Convert chat to PDF



Edit Image

Ask Gemini

Flash

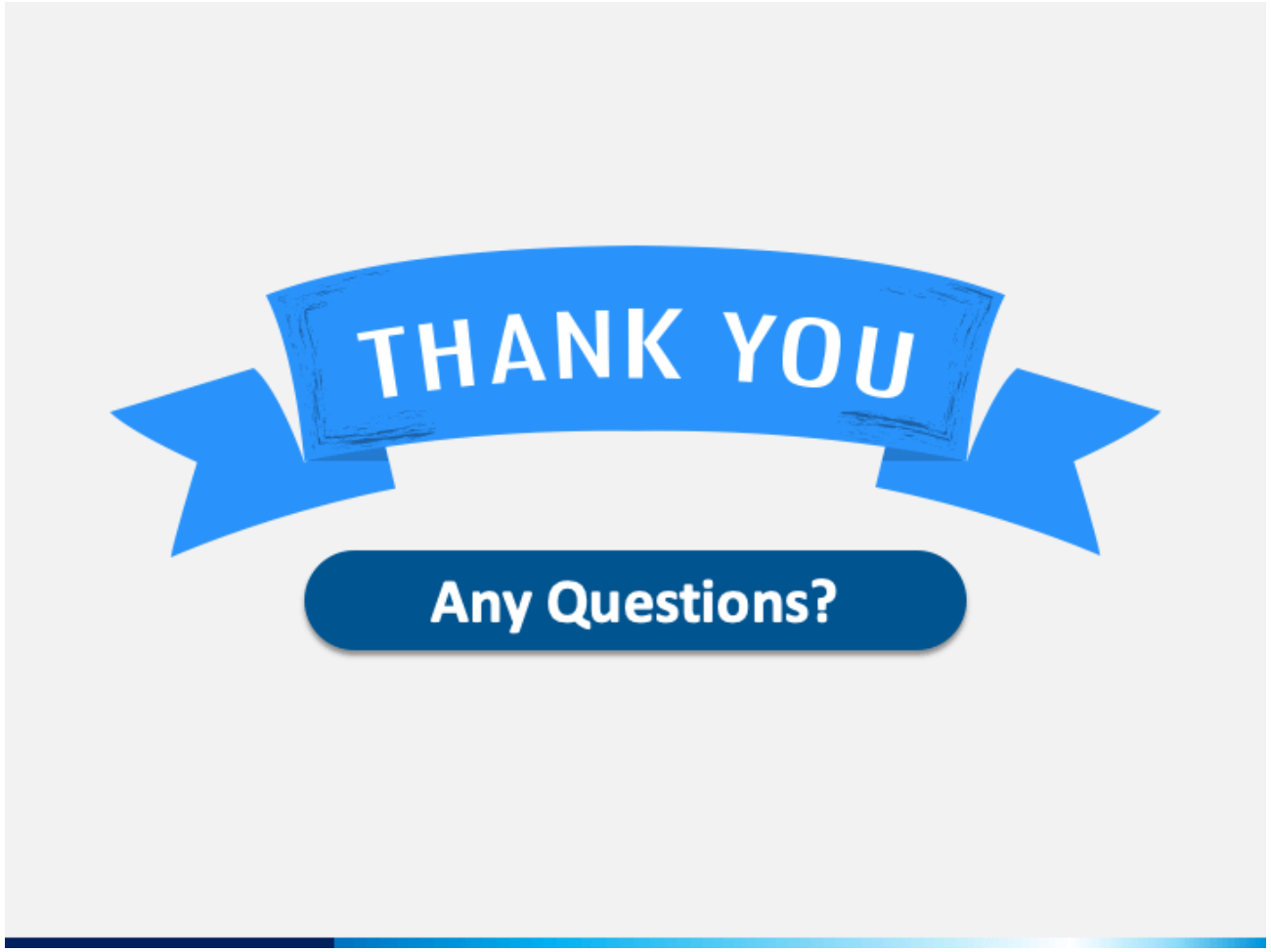
Gemini is AI and can make mistakes.

REFLECTION

Key takeaways

- A single missing control can cascade into total system loss.
- Detection tuned too late is nearly as costly as no detection at all.
- Backup strategy is a security control, not just an IT convenience.

One phishing click, five missing controls, nine hours to total loss and a clear, evidence-based path back.


THANK YOU
Any Questions?